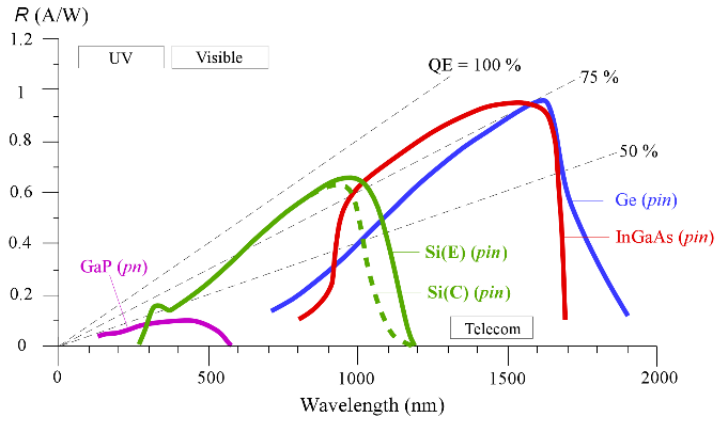
Marmara Üniversitesi
Electrical and Electronics Engineering Department
Fall 2023
Optoelectronics - Homework (2) – Due December 15, 2023

Part I: Photodiode Characterization

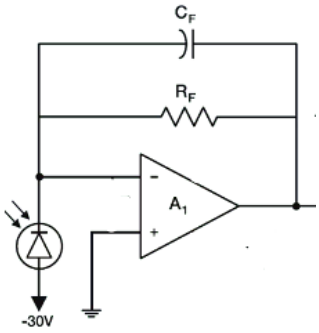
Consider the commercial InGaAs, Ge, Si (Enhanced and Conventional) pin photodiodes. The responsivity curves are given in the figure below.

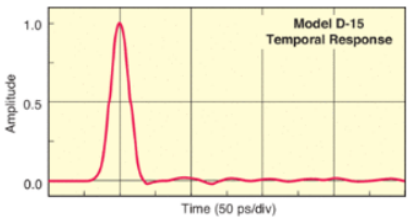


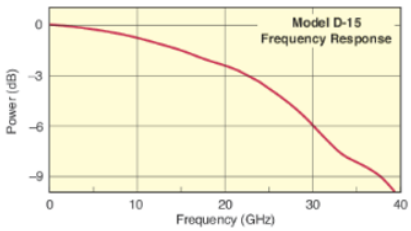
1) For each photodiode, find the optical power at 1550 nm that will give a photocurrent of 10 nA. Comment on the answers.
2) What would be the QE for each photodiode in part (1)? Comment on the answers.
3) For each photodiode, for the same optical power in part (1), find the photocurrent at 1064 nm.
4) Plot the QE of the InGaAs pin photodiode versus wavelength. Comment on the figure.

Part II: Photodiodes Circuits

Design the amplifier circuit below to produce the required filtering of the given signal.

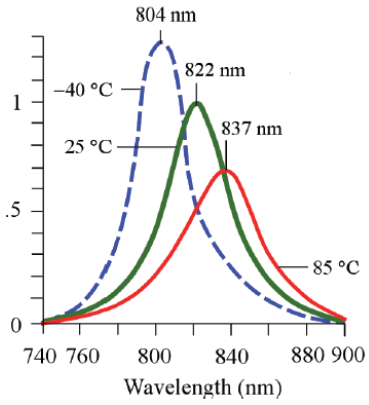






Part II: LEDs

The emission spectrum for an AlGaAs IR LED is given in the figure below for three different temperatures. Verify the peak wavelength and spectral linewidth vs. *T* relations for this case.



Upload your solution file (pdf only) on the link below.
https://www.dropbox.com/request/x3BB4Hoq1aZJJbTYs6gd